

Random Signal Theory–EEE 554

Lecture Notes 2

Classes and Moments of Random Variables

[Reading: Kay, Sections 5.5, 5.7, 10.5, 10.7, 6.3-6.8, 11.3-11.9]

Discrete RV Classes

probability mass function (PMF) [Kay, Section 5.5]

Bernoulli RV: $X \sim \text{Ber}(p)$

models one success with probability p and one failure with probability $q = 1 - p$

$$\text{PMF: } f_X(x) = \begin{cases} 1 - p = q, & x = 0 \\ p, & x = 1 \end{cases}$$

Poisson RV: $X \sim \text{Pois}(\lambda)$

models events occurring in a fixed interval of time at a known constant rate λ

$$f_X(x) = \sum_{k=0}^{\infty} \left(\frac{e^{-\lambda} \lambda^k}{k!} \right) \delta(x - k)$$

Example: Number of packets arriving at a node in a network system at a rate $\lambda > 0$

Binomial RV: $X \sim \text{bin}(N, p)$

Models n successes, each with probability p , out of N trials

$$\text{PMF: } f_X(x) = \sum_{n=0}^N \Pr(x = n) \delta(x - n) = \sum_{n=0}^N \left[\binom{N}{n} p^n q^{N-n} \right] \delta(x - n)$$

$$\text{where } \binom{N}{n} = \frac{N!}{n!(N-n)!}$$

Since area under PMF must be one, $\int_{-\infty}^{\infty} f_X(x) dx = 1$

$$\begin{aligned} \int_{-\infty}^{\infty} \left(\sum_{k=0}^N \Pr(x = k) \delta(x - k) \right) dx &= \int_{-\infty}^{\infty} \left(\sum_{k=0}^N \binom{N}{k} p^k q^{N-k} \delta(x - k) \right) dx \\ &= \sum_{k=0}^N \binom{N}{k} p^k q^{N-k} \int_{-\infty}^{\infty} \delta(x - k) dx = \sum_{k=0}^N \binom{N}{k} p^k q^{N-k} = 1 \end{aligned}$$

since $\int_{-\infty}^{\infty} \delta(x - k) dx = 1$ and considers all successes and failures

Continuous RV Classes

Probability density function (PDF), cumulative density function (CDF) [Kay, Section 10.5]

Uniform RV: $X \sim \mathcal{U}(x_1, x_2)$

Models events that are equally likely to occur between x_1 and x_2

$$\text{PDF: } f_X(x) = \begin{cases} \frac{1}{x_2 - x_1}, & x_1 < x < x_2 \\ 0, & \text{otherwise} \end{cases}, \quad \text{CDF: } F_X(x) = \begin{cases} 0, & x \leq x_1 \\ \frac{(x - x_1)}{x_2 - x_1}, & x_1 < x < x_2 \\ 1, & x \geq x_2 \end{cases}$$

Exponential RV: $X \sim \exp(\lambda)$

Models events that occur independently at a constant average rate λ

$$\text{PDF: } f_X(x) = \lambda e^{-\lambda x} u(x), \quad \text{CDF: } F_X(x) = (1 - e^{-\lambda x}) u(x)$$

where $u(x)$ is the unit step function

Common examples: arrivals at rate λ or growth at rate λ

Rayleigh RV: with parameter σ

Models magnitude affected by direction (e.g., models effects of fading channels)

$$\text{PDF: } f_X(x) = \frac{x}{\sigma^2} \exp\left\{\left(-\frac{x^2}{2\sigma^2}\right)\right\} u(x)$$

Other common CRVs: Gamma, χ^2 , Beta, Laplace, Cauchy, Wishart

Gaussian or normal RV: $X \sim \mathcal{N}(\mu, \sigma^2)$

Models linear combinations of different, independent observations
(converge to normal RV)

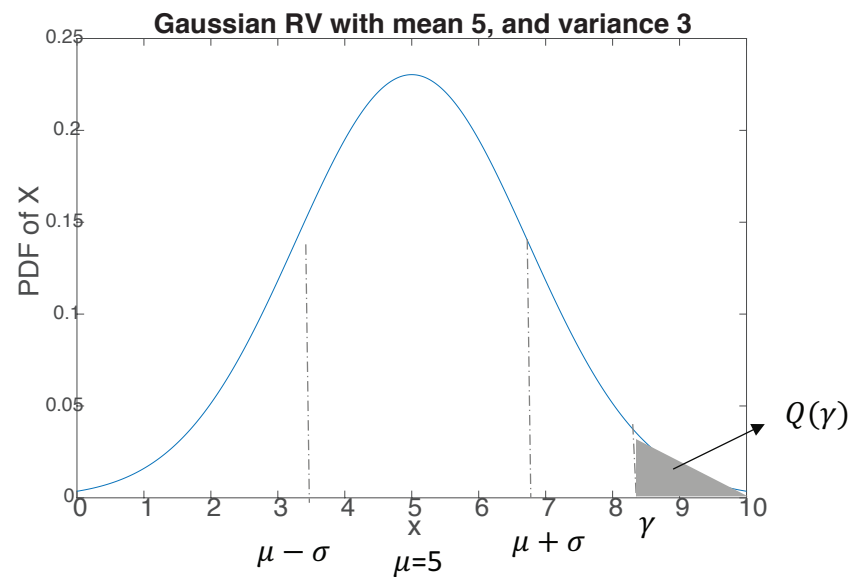
$$\text{PDF: } f_X(x) = \frac{1}{\sqrt{2\pi} \sigma} \exp\left\{\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right)\right\}, \forall x$$

If $Y = \sum_{i=1}^M X_i$ is the sum of M independent and identically distributed (iid) Gaussian RVs

$X_i \sim \mathcal{N}(\mu, \sigma^2)$, $i = 1, \dots, M$, then Y is also Gaussian $Y \sim \mathcal{N}(M\mu, M\sigma^2)$

Example: $X \sim \mathcal{N}(5, 3)$ simulated in MATLAB

```
N=1000; xmin=0; xmax=10; x=linspace(xmin,xmax,N);  
mu=5; var=3; fx=normpdf(x,mu,sqrt(var));  
plot(x,fx); xlabel('x'); ylabel('PDF of X');  
title(sprintf('Gaussian RV with mean %d, and variance%0.5g',mu,var));
```



Note: $\sim 2/3$ of area is within $\mu - \sigma$ and $\mu + \sigma$

CDF of a Gaussian RV cannot be expressed in closed form

For normalized Gaussian RV $Y \sim \mathcal{N}(0, 1)$, CDF is tabulated or computed in MATLAB as the Q-function or right-tail of the bell curve (see shaded area on plot in previous page)

$$Q(\gamma) = 1 - \Phi(\gamma) = 1 - F_Y(\gamma) = \int_{-\infty}^{\gamma} f_X(x) \, dx$$

[Kay, MATLAB code, page 341]

Given $X \sim \mathcal{N}(\mu, \sigma^2)$, $F_X(x) = \Phi\left(\frac{x - \mu}{\sigma}\right) = 1 - Q\left(\frac{x - \mu}{\sigma}\right)$

In MATLAB, the function `erfc.m` is used instead of the Q-function

Relation between functions: $Q(x) = 0.5 \operatorname{erfc}\left(\frac{x}{\sqrt{2}}\right)$

	Values	PDF	$E[X]$	$\text{var}(X)$	$\phi_X(\omega)$
Uniform	$a < x < b$	$\frac{1}{b-a}$	$\frac{1}{2}(a+b)$	$\frac{(b-a)^2}{12}$	$\frac{\exp(j\omega b) - \exp(j\omega a)}{j\omega(b-a)}$
Exponential	$x \geq 0$	$\lambda \exp(-\lambda x)$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda - j\omega}$
Gaussian	$-\infty < x < \infty$	$\frac{\exp[-(1/(2\sigma^2))(x-\mu)^2]}{\sqrt{2\pi\sigma^2}}$	μ	σ^2	$\exp[j\omega\mu - \sigma^2\omega^2/2]$
Laplacian	$-\infty < x < \infty$	$\frac{1}{\sqrt{2\sigma^2}} \exp(-\sqrt{2/\sigma^2} x)$	0	σ^2	$\frac{2/\sigma^2}{\omega^2 + 2/\sigma^2}$
Gamma	$x \geq 0$	$\frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} \exp(-\lambda x)$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$	$\frac{1}{(1-j\omega/\lambda)^\alpha}$
Rayleigh	$x \geq 0$	$\frac{x}{\sigma^2} \exp[-x^2/(2\sigma^2)]$	$\sqrt{\frac{\pi\sigma^2}{2}}$	$(2-\pi/2)\sigma^2$	[Johnson et al 1994]

Table 11.1: Properties of continuous random variables.